

Fort Saskatchewan Path₂Zero Project - UPI

Standard Girt Design

FS-D000-CAL-20004

P.Eng Stamp	Permit to Practice Stamp
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Checking Level: 2B

Doc Rev	Revision Description/ Status	Date Issued	Fluor Originator	Checker	Approval
				Fluor Engineering Checker	Fluor Discipline Lead
0	IFI	27-Oct-2023	A. WALL	A. SHARIF	K. TULLU

Risk Notice

Does this Calculation have any assumptions or risk items that need to be followed up with?

☐ Yes ☒ No

If Yes, anticipated date the information will be available: _____

Followed up by: _____ Date: _____ Signature: _____

Client: DOW CHEMICAL CANADA ULC
 Project No: P2ZU

Revision History

Description of revision, especially to original calculation

Rev No	Originator (Name)	Checker (Name)	Revision Date	Revision Description
0	Alex Wall	Alexander Sharif	27-Oct-2023	IFI - Issued for Information

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Client: DOW CHEMICAL CANADA ULC
 Project No: P2ZU

1. RISK AND UNCERTAINTY LOG

The purpose of the table outlines the uncertainties, mitigation strategy, remaining risk at the time of design.
 Assumptions to be re-validated once the appropriate information is made available.

Seq # (PDR # if avail)	Uncertainty	Mitigation	Remaining Risk	Resolved (Y/N)	Resolution (include approx dat

Client: DOW CHEMICAL CANADA ULC

Project No: P2ZU

2. COMPUTER FILE INDEX

Title: Standard Girt Design
Calculation Number: FS-D000-CAL-20004

The purpose of this form is to keep track of calculation computer files and to assign filenames using a standard format. This will assist in organizing data and in retrieving data at a later date. The file naming scheme shown below should be used.

Include this form with each set of calculations which has computer files. List all data files that were used to complete the calculation. Filenames shall be as follows:

FS-AXXX-ZZZ-DDNNN_R.EXT

Where:

- FS = Project Code
- A = Page Size (A - 22x34, B - 17x23, C - 17x11, D - 8.5x11, F - Other)
- XXX = Area / Unit Number
- ZZZ = Document Type (CAL - Calculations)
- DD = Discipline Code (21 - Civil / 20 - Structural)
- NNN = Sequence Number
- R = Revision Number
- EXT = File extension (eg: xls,xlsx, r3d, docx, pdf, etc.)

FILENAME	DESCRIPTION	DIRECTORY LOCATION
FS-D000-CAL-20004_0.xls	Native Excel Calculation File	\\215_STRUCTURAL\215E_ENGINEERING_DESIGN_DATA\215E_014_R_Proj_Calculations_Sketches\WER03\FS-D000-CAL-20004\
FS-D000-CAL-20004_0.pdf	PDF Output	

3. DESCRIPTION OF CALCULATION

This calculation pertains to site-wide standardization of girt design to assist in selection during detailed design of the various buildings

Standardized girt design will consider columns spaced at 15' (4.572 m), 20' (6.096 m), 25' (7.62 m) and 30' (9.144 m), and typical vertical girt spacing of 5' (1.524 m).

For the purpose of wind pressure, a maximum wind-band height of 40' (12.192 m) is considered.

Buildings taller than this must consider additional wind pressure acting on the girts

3.1 Description of the Structure

Wide-flange beams are considered for girts. No sag rods are considered. Bolted connections to column are designed by others.

A786/A786M minimum yield strength 35 ksi (240 MPa) (UNO)
Grating and Stair Tread – Welded steel grating w/ 1 1/4" x 3/16" (32x4.8)
serrated bearing bars at 1 3/16" (30 mm) centers

Structural Bolts (UNO): High Strength bolts – 3/4" Dia ASTM F3125, Grade A325, Type 1(UNO)

Connection plate: Min. fillet weld size shall be 1/4" (6mm)
All gusset plates and connection plate materials shall be min. 3/8"
(10mm) thick

4.6 Software

The following outlines the list of computer software used for analysis. Refer to the Computer File Index for files:

Microsoft Excel	365
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5. ANALYSIS AND DESIGN PHILOSOPHY

As the girts only support dead load (self-weight) and wind loading, the first step is to determine the magnitude of wind pressure acting on the tributary area of each girt. Once the appropriate loading cases are determined the required girt section will be governed by column spacing. Due to the relatively simple connections, assumed to be pinned, the girt designs will be governed by strong axis bending (wind) and weak-axis bending (self-weight).

Wind loading transferred into the girts will be a function of outside pressure as well as internal suction. Total wind pressure is determined using 2015 NBC Structural commentary I static procedure considering worst-case exterior wall geometry.

6. LOAD DEVELOPMENT

Only BLC 1 and BLC 17 are considered applicable for Girt design. Seismic is not considered to govern due to relatively small proportion of dead load.

Basic Load Cases: The table below shows the basic load cases that were specified from the job data book Jdb2.20

Table 4-1 Basic Load Cases

Basic Load Case	Symbol	Description	Applied
BLC 1	SW	Self-weight, fireproofing, and misc. other dead load	Y
BLC 2	PO	Piping Operating Load	N
BLC 3	PE	Piping Empty Load	N
BLC 4	ET	Pipe / Equipment Test Load - "Content Weight Alone"	N
BLC 5	F	Pipe Friction Load	N
BLC 6	AT	Pipe Anchor Thermal	N
BLC 7	AF	Pipe Anchor Friction	N
BLC 8	AS	Pipe Anchor Seismic	N
BLC 9	GL	Guide Load	N
BLC 10	SG	Slug or Upset Loads	N
BLC 11	EL	Electrical	N
BLC 12	EO	Equipment Operating	N
BLC 13	EE	Equipment Empty	N
BLC 14	EM	Equipment Maintenance Load (ex. Bundle Pull)	N
BLC 15	L	Live Load	N
BLC 16	S	Snow Load	N
BLC 17	WT	Wind Transverse	Y
BLC 18	WL	Wind Longitudinal	N
BLC 19	STOpe	Seismic Trans Operating	N
BLC 20	STEmp	Seismic Trans Empty	N
BLC 21	SLOpe	Seismic Long Operating	N
BLC 22	SLEmp	Seismic Long Empty	N
BLC 23	Di	Atmospheric Ice (Di), if applicable	N
BLC 24	TH,Stcon	Thermal Effects Contraction	N
BLC 25	TH,Stexp	Thermal Effects Expansion	N
BLC 26	-	Additional BLC, as required	-
Transportation Load Cases			
BLC 28	T1	Self-weight Longitudinal	N
BLC 29	T2	Self-weight Transverse	N
BLC 30	T3	Pipe Empty Longitudinal	N
BLC 31	T4	Pipe Empty Transverse	N
BLC 32	T5	Equipment Empty Longitudinal	N
BLC 33	T6	Equipment Empty Transverse	N
BLC 34	T7	Electrical Longitudinal	N
BLC 35	T8	Electrical Transverse	N
BLC 36	T9	Snow Longitudinal	N
BLC 37	T10	Snow Transverse	N

6.1 BLC 1 - Self-weight

Self-weight will include the girt as well as any cladding clipped to the girt beams. A 15% factor is considered for connection weight

Table 6-1 Girt Section SW Summary

Metric Designation	Imperial Designation	Section Unit Mass (kg/m)	Self Weight (kN/m)	Design Dead Load* (kN/m)
W150x22	W6x15	22	0.216	0.248
W200x19	W8x13	19	0.186	0.214
W200x21	W8x14	21	0.206	0.237
W200x22	W8x15	22	0.216	0.248
W200x27	W8x18	27	0.265	0.305
W200x31	W8x21	31	0.304	0.350
W200x36	W8x24	36	0.353	0.406
W200x42	W8x28	42	0.412	0.474
W200x46	W8x31	46	0.451	0.519
W200x52	W8x35	52	0.510	0.587
W250x18	W10x12	18	0.177	0.203
W250x22	W10x15	22	0.216	0.248
W250x25	W10x17	25	0.245	0.282
W250x24	W10x16	24	0.235	0.271
W250x28	W10x19	28	0.275	0.316
W250x33	W10x22	33	0.324	0.372
W250x39	W10x26	39	0.383	0.440
W250x45	W10x30	45	0.441	0.508
W250x49	W10x33	49	0.481	0.553
W250x58	W10x39	58	0.569	0.654

*Design dead load includes the 15% factor considered for connection weight

6.1.1 Cladding Self-weight

Cladding used on the project is expected to vary, so preliminary cladding weight is based on exterior wall assembly system W-3 used in BLDG-77C (Paint Shop).

- W-3** - PREFINISHED GALVANIZED METAL PANEL, 22 GAUGE, MORIN (KINGSPAN) EXPOSED VB-31-S PROFILE "A" METAL CLADDING OR VICWEST CL7040. (TO BE SIMILAR TO CLADDING PROFILE OF THE EXISTING BUILDINGS ON-SITE)
- STRUCTURAL STEEL GIRTS
 - 8" THICK CONTINUOUS R24 MINERAL WOOL INSULATION (MIN DENSITY 96kg/m³) INSTALLED WITH JOINTS STAGGERED
 - AIR/VAPOR RETARDER
 - 5/8" GYPSUM BOARD (MOISTURE AND MOLD RESISTANT REQUIRED FOR ROOMS 105 AND 106)

Unit weight of gauge 22 (0.7mm) Vicwest CL7040 $q_{cladding}$ = 0.08 kPa

Unit weight of insulation (min density 96kg/m³, 8" thk) q_{insul} = 0.191 kPa

Unit weight of gypsum board, 0.08kN/m² per 10mm thk q_{gypsum} = 0.127 kPa

Assumed cladding dead load q_c = 0.40 kPa

Design Cladding Dead Load (5' height, includes 15% for connections) w_c = 0.70 kN/m

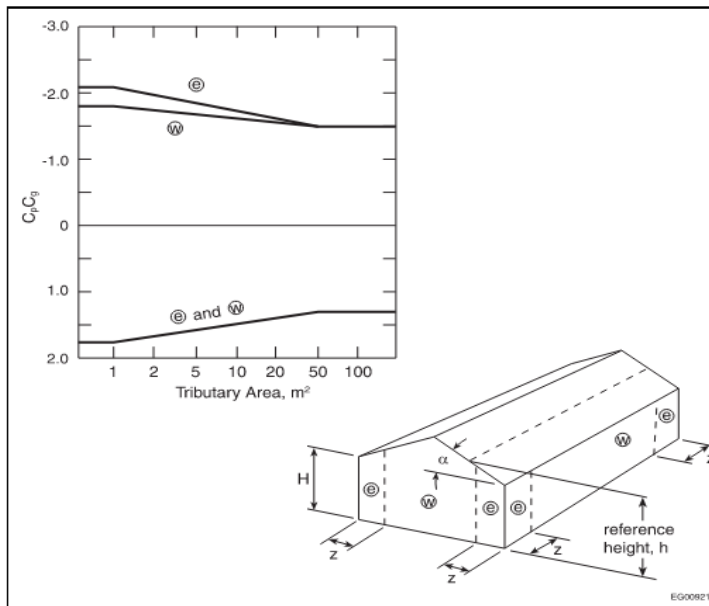
6.2 BLC 17 - Wind Loading

Wind load calculations are as per NBC (AE) 2019. Reference calculation site parameters are taken from Doc. FS2Z-0001-33. Reference wind band height taken as 35' (10.67 m) based on the tallest anticipated building.

Building height is assumed to be < 20 m and less than the smaller plan dimension.

6.2.1 Wind Loading Exterior Pressure / Suction

Figure 4.1.7.6.-B
External peak values of $C_p C_g$ on individual walls for the design of cladding and secondary structural members
Forming Part of Sentence 4.1.7.6.(3)



Notes to Figure 4.1.7.6.-B:

- (1) These coefficients apply for any roof slope, α .
- (2) End-zone width z is the lesser of 10% of the least horizontal dimension and 40% of height, H , but not less than 4% of the least horizontal dimension or 1 m.
- (3) Combinations of external and internal pressures must be evaluated to obtain the most severe loading.
- (4) Positive coefficients denote forces toward the surface, whereas negative coefficients denote forces away from the surface. Each structural element must be designed to withstand forces of both signs.
- (5) Pressure coefficients generally apply for facades with architectural features; however, where vertical ribs deeper than 1 m are placed on a facade, a local $C_p C_g$ of -2.8 applies to zone e.

ULS wind importance factor	I_w	=	1	Normal considered for secondary member
Reference velocity pressure	q	=	0.43 kPa	
Reference height	h	=	10.67 m	
Open terrain exposure factor, varies by height	C_e	=	1.013	$(h/10)^{0.2}$
Topographic factor	C_t	=	1	
External peak values of $C_p C_g$ Acting Inwards	$C_p C_{g,in}$	=	1.75	(From 4.1.7.6.-B)
External peak values of $C_p C_g$ Acting Outwards	$C_p C_{g,out}$	=	2.125	(From 4.1.7.6.-B)
ULS wind pressure for open structure	p	=	$I_w q C_e C_t C_g C_p$	
ULS Exterior wind pressure acting Inwards	$p_{ext,in}$	=	0.76 kPa	
ULS Exterior wind pressure acting Outwards	$p_{ext,out}$	=	0.93 kPa	
Exterior Wind Load Acting Inwards	$W_{WL,ext,in}$	=	1.16 kN/m	(Total acting over 5' girt tributary height)
Exterior Wind Load Acting Outwards	$W_{WL,ext,out}$	=	1.41 kN/m	

6.2.2 Wind Loading Interior Pressure / Suction

4.1.7.7. Internal Pressure Coefficient

- 1) The internal pressure coefficient, C_{pi} , shall be as prescribed in Table 4.1.7.7.

Table 4.1.7.7.
Internal Pressure Coefficients
Forming Part of Sentence 4.1.7.7.(1)

Building Openings	Values for C_{pi}
Uniformly distributed small openings amounting to less than 0.1% of the total surface area of the building	-0.15 to 0.0
Non-uniformly distributed openings of which none is significant or significant openings that are wind-resistant and closed during storms	-0.45 to +0.30
Large openings likely to remain open during storms	-0.70 to +0.70

ULS wind importance factor	I_w	=	1
Reference velocity pressure	q	=	0.43 kPa
Open terrain exposure factor, varies by height	C_e	=	1.013 $(h/10)^{0.2}$
Topographic factor	C_t	=	1
Internal Gust Effect Factor	C_g	=	2 (From 4.1.7.3. Sentence 8)
Internal Pressure Coefficient Acting Inwards	$C_{p,int,in}$	=	0.45
Internal Pressure Coefficient Acting Outwards	$C_{p,int,out}$	=	0.3
Reference height	h	=	10.67 m
ULS wind pressure for open structure	p	=	$I_w q C_e C_t C_g C_p$
ULS Interior wind pressure acting Inwards	$p_{int,in}$	=	0.39 kPa
ULS Interior wind pressure acting Outwards	$p_{int,out}$	=	0.26 kPa

Design wind load acting on the girt is a function of girt spacing. Assume 5' (1.524 m)

Design Wind Load	$W_{WL,int,in/ext}$	=	0.60 kN/m
Design Wind Load	$W_{WL,int,in/ext}$	=	0.40 kN/m

The internal pressure coefficient is taken as the worst-case value of 0.7. This value is reversible, therefore the design wind load is the same for inward and outward loading.

6.2.3 Total Design Wind Loading

Design Wind Load Acting Inwards	$W_{WL,int}$	=	1.76 kN/m
Design Wind Load Acting Outwards	$W_{WL,out}$	=	1.81 kN/m

Wind acting inwards and outwards will result in similar bending on the girt, however the direction of wind is important in determining structural unity as wind acting inwards allows the compression flange to be fully laterally supported, whereas wind acting outwards must consider the beam laterally unsupported over the distance between columns. Because the wind acting outward is a higher magnitude, and also a more critical design scenario, the wind acting inwards will not be considered further.

7. LOAD COMBINATIONS

Applicable load combinations for Girt design include the following:

- 1 1.4DL
- 2 1.25DL + 1.4WL

Dead loads will induce weak-axis bending in the girts, while wind loading will induce strong-axis bending. The governing loading scenario will include bi-axial bending and bi-axial shear.

Table 7-1 Girt SLS Serviceability and ULS Factored Loading

Metric Designation	Imperial Designation	SLS Loading		ULS Loading		
		1.0DL (kN/m)	0.75WL (kN/m)	1.4DL (kN/m)	1.25DL (kN/m)	1.4WL (kN/m)
W150x22	W6x15	0.947	1.36	1.33	1.18	2.53
W200x19	W8x13	0.913	1.36	1.28	1.14	2.53
W200x21	W8x14	0.935	1.36	1.31	1.17	2.53
W200x22	W8x15	0.947	1.36	1.33	1.18	2.53
W200x27	W8x18	1.003	1.36	1.40	1.25	2.53
W200x31	W8x21	1.048	1.36	1.47	1.31	2.53
W200x36	W8x24	1.105	1.36	1.55	1.38	2.53
W200x42	W8x28	1.172	1.36	1.64	1.47	2.53
W200x46	W8x31	1.217	1.36	1.70	1.52	2.53
W200x52	W8x35	1.285	1.36	1.80	1.61	2.53
W250x18	W10x12	0.902	1.36	1.26	1.13	2.53
W250x22	W10x15	0.947	1.36	1.33	1.18	2.53
W250x25	W10x17	0.980	1.36	1.37	1.23	2.53
W250x24	W10x16	0.969	1.36	1.36	1.21	2.53
W250x28	W10x19	1.014	1.36	1.42	1.27	2.53
W250x33	W10x22	1.071	1.36	1.50	1.34	2.53
W250x39	W10x26	1.138	1.36	1.59	1.42	2.53
W250x45	W10x30	1.206	1.36	1.69	1.51	2.53
W250x49	W10x33	1.251	1.36	1.75	1.56	2.53
W250x58	W10x39	1.353	1.36	1.89	1.69	2.53

Because the peak wind loading is acting outwards, the cladding cannot be used as bracing along the length of the compression flange, therefore the effective length of the girt is taken as the distance between columns.

8. ANALYSIS AND RESULTS

Analysis for the shear and bending capacity of the girt sections is performed as detailed in CSA S16-14. The girts experience bi-axial bending and shear as well as local bending about the web caused by eccentric dead load. The beams are considered laterally unsupported over the entire length (column spacing).

Table 8-1 Girt Section Overall Summary

Section	Columns @ 15'			Columns @ 20'			Columns @ 25'			Columns @ 30'		
	UC _{Mx,y}	UC _{V,15'}	Δ/L Ratio	UC _{Mx,y}	UC _{V,15'}	Δ/L Ratio	UC _{Mx,y}	UC _{V,15'}	Δ/L Ratio	UC _{Mx,y}	UC _{V,15'}	Δ/L Ratio
W150x22	0.36	0.05	L / 1433	0.72	0.06	L / 605	1.32	0.08	L / 310	2.17	0.09	L / 179
W200x19	0.72	0.03	L / 1966	1.51	0.05	L / 830	2.73	0.06	L / 425	4.44	0.07	L / 246
W200x21	0.44	0.04	L / 2346	0.94	0.05	L / 990	1.73	0.07	L / 507	2.85	0.08	L / 293
W200x22	0.57	0.03	L / 2369	1.19	0.04	L / 1000	2.11	0.05	L / 512	3.41	0.06	L / 296
W200x27	0.33	0.03	L / 3056	0.69	0.05	L / 1289	1.23	0.06	L / 660	1.99	0.07	L / 382
W200x31	0.27	0.03	L / 3720	0.54	0.04	L / 1569	0.95	0.05	L / 803	1.52	0.06	L / 465
W200x36	0.19	0.03	L / 4075	0.36	0.05	L / 1719	0.64	0.06	L / 880	1.03	0.07	L / 509
W200x42	0.17	0.03	L / 4845	0.31	0.04	L / 2044	0.54	0.05	L / 1047	0.85	0.06	L / 606
W200x46	0.13	0.03	L / 5378	0.25	0.04	L / 2269	0.41	0.05	L / 1162	0.63	0.06	L / 672
W200x52	0.11	0.03	L / 6243	0.21	0.04	L / 2634	0.35	0.05	L / 1348	0.54	0.06	L / 780
W250x18	0.89	0.03	L / 2654	1.93	0.04	L / 1119	3.54	0.06	L / 573	5.86	0.07	L / 332
W250x22	0.65	0.03	L / 3424	1.37	0.04	L / 1444	2.47	0.05	L / 739	4.02	0.06	L / 428
W250x25	0.53	0.03	L / 4051	1.10	0.04	L / 1709	1.97	0.04	L / 875	3.17	0.05	L / 506
W250x24	0.35	0.03	L / 4111	0.75	0.04	L / 1734	1.39	0.05	L / 888	2.31	0.07	L / 514
W250x28	0.44	0.03	L / 4738	0.91	0.03	L / 1999	1.60	0.04	L / 1024	2.56	0.05	L / 592
W250x33	0.25	0.03	L / 5793	0.51	0.04	L / 2444	0.91	0.05	L / 1251	1.47	0.05	L / 724
W250x39	0.20	0.03	L / 7119	0.41	0.03	L / 3004	0.71	0.04	L / 1538	1.13	0.05	L / 890
W250x45	0.18	0.02	L / 8423	0.34	0.03	L / 3553	0.59	0.04	L / 1819	0.93	0.04	L / 1053
W250x49	0.13	0.02	L / 8363	0.23	0.03	L / 3528	0.39	0.04	L / 1806	0.62	0.05	L / 1045
W250x58	0.10	0.02	L / 10342	0.19	0.03	L / 4363	0.32	0.04	L / 2234	0.50	0.05	L / 1293

UC_{Mx,y} Considers bi-axial bending for load combination 2: 1.25DL + 1.4WL

UC_{V,15'} Considers maximum shear resulting from load combination 2: 1.25DL + 1.4WL

Δ/L Ratio Considers maximum SLS deflection from wind loading. Maximum allowable SLS deflection is L/180

Summarized results are based on 5' vertical spacing of girts. Tighter vertical spacing will reduce overall loading, and proportionally reduce deflections and required section size.

8.1 Bending and Shear Unity Checks

8.1.1 Material Property Input

350W Yield Strength	f_y	=	350	MPa
350W Ultimate Strength	f_u	=	450	MPa
Elastic Modulus	E	=	200000	MPa
Shear Modulus	G	=	77000	MPa
Structural Steel Resistance Factor	ϕ_s	=	0.9	

8.1.2 Section Geometric Properties

Table 8-2 Wide Flange Beam Geometric Properties

Metric Designation	I_x (10^6 mm^4)	I_y (10^6 mm^4)	Z_x (10^3 mm^3)	Z_y (10^3 mm^3)	w (mm)	Depth (mm)	Class	Flange Thk t (mm)	J (10^3 mm^4)	C_w (10^9 mm^6)
W150x22	12.1	3.87	176	77.6	6	152	4	6.6	42.0	20.4
W200x19	16.6	1.15	187	35.6	6	203	2	6.5	36.2	11.1
W200x21	19.8	2.51	217	58	5	203	3	6.4	35.6	24.3
W200x22	20	1.42	222	43.7	6	206	1	8.0	56.6	13.9
W200x27	25.8	3.3	279	76.1	6	207	2	8.4	71.3	32.5
W200x31	31.4	4.1	335	93.8	6	210	1	10.2	119.0	40.9
W200x36	34.4	7.64	379	141	6	201	2	10.2	145.0	69.6
W200x42	40.9	9	445	165	7	205	1	11.8	222.0	84.0
W200x46	45.4	15.3	495	229	7	203	3	11.0	220.0	141.0
W200x52	52.7	17.8	569	266	8	206	2	12.6	323.0	167.0
W250x18	22.4	0.913	207	28.6	5	251	3	5.3	22.4	13.8
W250x22	28.9	1.23	263	38.1	6	254	1	6.9	43.4	18.7
W250x25	34.2	1.49	307	46.2	6	257	1	8.4	65.2	23.0
W250x24	34.7	3.26	307	69	5	253	4	6.4	39.7	49.5
W250x28	40	1.78	353	54.7	6	260	1	10.0	96.7	27.7
W250x33	48.9	4.73	424	99.5	6	258	2	9.1	98.5	73.2
W250x39	60.1	5.94	513	124	7	262	1	11.2	169.0	93.4
W250x45	71.1	7.03	602	146	8	266	1	13.0	261.0	113.0
W250x49	70.6	15.1	633	228	7	247	3	11.0	241.0	211.0
W250x58	87.3	18.8	770	283	8	252	1	13.5	409.0	268.0

8.1.2 Wide Flange Bending

Section strong-axis bending capacity is computed according to 13.6 for each section size for column spacing of 15', 20', 25' and 30'

13.6 Bending — Laterally unsupported members

13.6.1 General construction

Where continuous lateral support is not provided to the compression flange of a member subjected to uniaxial strong axis bending, the factored moment resistance, M_r , of a segment between effective brace points shall be determined as follows:

a) For doubly symmetric Class 1 and 2 sections, except closed square and circular sections:

i) when $M_u > 0.67M_p$:

$$M_r = 1.15\phi M_p \left[1 - \frac{0.28M_p}{M_u} \right] \leq \phi M_p$$

ii) when $M_u \leq 0.67M_p$:

$$M_r = \phi M_u$$

where the critical elastic moment of the unbraced segment, M_u , is given by

$$M_u = \frac{\omega_2 \pi}{L} \sqrt{EI_y GJ + \left(\frac{\pi E}{L} \right)^2 I_y C_w}$$

ω_2 = coefficient to account for increased moment resistance of a laterally unsupported doubly symmetric beam segment when subject to a moment gradient

$$= \frac{4M_{max}}{\sqrt{M_{max}^2 + 4M_a^2 + 7M_b^2 + 4M_c^2}} \leq 2.5$$

ω_2 can be simplified, as $M_a = M_c = 3/32\omega_1^2$. $M_b = M_{max} = 1/8\omega_1^2$

$$\omega_2 = 1.13$$

Table 8-3 Wide Flange Beam Strong-Axis Factored Bending Capacity

Metric Designation	$M_{p,x}$ or $M_{y,x}$ (kNm)	$M_{u,x}$ 15ft (kNm)	$M_{r,x}$ 15ft (kNm)	$M_{u,x}$ 20ft (kNm)	$M_{r,x}$ 20ft (kNm)	$M_{u,x}$ 25ft (kNm)	$M_{r,x}$ 25ft (kNm)	$M_{u,x}$ 30ft (kNm)	$M_{r,x}$ 30ft (kNm)
W150x22	51.3	49.1	37.6	33.7	30.3	25.7	23.1	20.8	18.8
W200x19	65.5	23.1	20.8	16.2	14.6	12.6	11.3	10.3	9.3
W200x21	68.3	39.1	35.2	26.2	23.6	19.7	17.8	15.9	14.3
W200x22	77.7	31.2	28.1	22.2	20.0	17.3	15.5	14.2	12.8
W200x27	97.7	58.4	52.6	40.2	36.2	30.8	27.7	25.0	22.5
W200x31	117.3	80.3	71.8	56.2	50.6	43.4	39.1	35.4	31.9
W200x36	132.7	128.0	97.4	87.9	79.1	67.1	60.4	54.4	48.9
W200x42	155.8	165.0	118.6	114.8	100.0	88.4	79.6	72.0	64.8
W200x46	156.8	236.5	132.2	159.4	117.6	120.3	103.1	96.8	87.1
W200x52	199.2	295.7	167.3	202.1	149.2	153.8	131.4	124.5	112.0
W250x18	62.7	18.3	16.5	12.4	11.1	9.3	8.4	7.5	6.8
W250x22	92.1	27.6	24.8	19.0	17.1	14.6	13.1	11.9	10.7
W250x25	107.5	36.0	32.4	25.1	22.6	19.4	17.5	15.8	14.2
W250x24	89.7	55.2	49.7	35.5	32.0	25.9	23.3	20.4	18.4
W250x28	123.6	46.5	41.9	32.9	29.6	25.5	22.9	20.9	18.8
W250x33	148.4	91.0	81.9	60.7	54.7	45.5	41.0	36.5	32.8
W250x39	179.6	125.2	111.2	85.2	76.7	64.7	58.2	52.3	47.0
W250x45	210.7	161.7	138.5	111.7	100.5	85.6	77.0	69.5	62.6
W250x49	200.2	265.0	163.4	174.8	140.8	130.0	117.0	103.6	93.3
W250x58	269.5	359.3	220.3	241.8	191.9	182.3	163.4	146.6	131.9

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For bending about the weak-axis, $M_p = F_y Z_y$, $M_u = 0.67M_p$ and $M_r = \Phi_s M_u$

Table 8-4 Wide Flange Beam Weak-Axis Factored Bending Capacity

Metric Designation	$M_{p,y}$ (kNm)	$M_{u,y}$ (kNm)	$M_{r,y}$ (kNm)
W150x22	27.2	18.2	16.4
W200x19	12.5	8.3	7.5
W200x21	20.3	13.6	12.2
W200x22	15.3	10.2	9.2
W200x27	26.6	17.8	16.1
W200x31	32.8	22.0	19.8
W200x36	49.4	33.1	29.8
W200x42	57.8	38.7	34.8
W200x46	80.2	53.7	48.3
W200x52	93.1	62.4	56.1
W250x18	10.0	6.7	6.0
W250x22	13.3	8.9	8.0
W250x25	16.2	10.8	9.8
W250x24	24.2	16.2	14.6
W250x28	19.1	12.8	11.5
W250x33	34.8	23.3	21.0
W250x39	43.4	29.1	26.2
W250x45	51.1	34.2	30.8
W250x49	79.8	53.5	48.1
W250x58	99.1	66.4	59.7

Load combination 1 (1.4DL) considers only self-weight of the girt and any cladding. Bending unity ratio is computed as $UC = M_{f,y}/M_{r,y}$

Table 8-5 ULS Factored Bending Moment and Unity Ratios - Case 1 - 1.4DL

Section	Columns @ 15'		Columns @ 20'		Columns @ 25'		Columns @ 30'	
	$M_{f,15'}$	$UC_{15'}$	$M_{f,20'}$	$UC_{20'}$	$M_{f,25'}$	$UC_{25'}$	$M_{f,30'}$	$UC_{30'}$
W150x22	3.5	0.21	6.2	0.38	9.6	0.59	13.9	0.85
W200x19	3.3	0.44	5.9	0.79	9.3	1.23	13.4	1.78
W200x21	3.4	0.28	6.1	0.50	9.5	0.78	13.7	1.12
W200x22	3.5	0.38	6.2	0.67	9.6	1.04	13.9	1.50
W200x27	3.7	0.23	6.5	0.41	10.2	0.63	14.7	0.91
W200x31	3.8	0.19	6.8	0.34	10.7	0.54	15.3	0.77
W200x36	4.0	0.14	7.2	0.24	11.2	0.38	16.2	0.54
W200x42	4.3	0.12	7.6	0.22	11.9	0.34	17.2	0.49
W200x46	4.5	0.09	7.9	0.16	12.4	0.26	17.8	0.37
W200x52	4.7	0.08	8.4	0.15	13.1	0.23	18.8	0.33
W250x18	3.3	0.55	5.9	0.97	9.2	1.52	13.2	2.19
W250x22	3.5	0.43	6.2	0.77	9.6	1.20	13.9	1.72
W250x25	3.6	0.37	6.4	0.65	10.0	1.02	14.3	1.47
W250x24	3.5	0.24	6.3	0.43	9.8	0.68	14.2	0.97
W250x28	3.7	0.32	6.6	0.57	10.3	0.89	14.8	1.29
W250x33	3.9	0.19	7.0	0.33	10.9	0.52	15.7	0.75
W250x39	4.2	0.16	7.4	0.28	11.6	0.44	16.7	0.64
W250x45	4.4	0.14	7.8	0.25	12.3	0.40	17.6	0.57
W250x49	4.6	0.10	8.1	0.17	12.7	0.26	18.3	0.38
W250x58	4.9	0.08	8.8	0.15	13.7	0.23	19.8	0.33

Load combination 2 (1.25DL+1.4WL) considers self-weight of the girt plus any cladding and wind loading. Bending unity ratio is computed as $UC = M_{f,x}/M_{r,x} + M_{f,y}/M_{r,y}$. This loading scenario is much more critical with respect to overall unity in comparison with case 1 - 1.4DL due to the bi-axial bending and higher magnitude of wind loading.

Table 8-6 ULS Factored Bending Moment and Unity Ratios - Case 2 - 1.25DL+1.4WL

Section	Columns @ 15'			Columns @ 20'			Columns @ 25'			Columns @ 30'		
	$M_{f,x,15'}$	$M_{f,y,15'}$	UC _{15'}	$M_{f,x,20'}$	$M_{f,y,20'}$	UC _{20'}	$M_{f,x,25'}$	$M_{f,y,25'}$	UC _{25'}	$M_{f,x,30'}$	$M_{f,y,30'}$	UC _{30'}
W150x22	6.6	3.1	0.36	11.8	5.5	0.72	18.4	8.6	1.32	26.5	12.4	2.17
W200x19	6.6	3.0	0.72	11.8	5.3	1.51	18.4	8.3	2.73	26.5	11.9	4.44
W200x21	6.6	3.1	0.44	11.8	5.4	0.94	18.4	8.5	1.73	26.5	12.2	2.85
W200x22	6.6	3.1	0.57	11.8	5.5	1.19	18.4	8.6	2.11	26.5	12.4	3.41
W200x27	6.6	3.3	0.33	11.8	5.8	0.69	18.4	9.1	1.23	26.5	13.1	1.99
W200x31	6.6	3.4	0.27	11.8	6.1	0.54	18.4	9.5	0.95	26.5	13.7	1.52
W200x36	6.6	3.6	0.19	11.8	6.4	0.36	18.4	10.0	0.64	26.5	14.4	1.03
W200x42	6.6	3.8	0.17	11.8	6.8	0.31	18.4	10.6	0.54	26.5	15.3	0.85
W200x46	6.6	4.0	0.13	11.8	7.1	0.25	18.4	11.0	0.41	26.5	15.9	0.63
W200x52	6.6	4.2	0.11	11.8	7.5	0.21	18.4	11.7	0.35	26.5	16.8	0.54
W250x18	6.6	2.9	0.89	11.8	5.2	1.93	18.4	8.2	3.54	26.5	11.8	5.86
W250x22	6.6	3.1	0.65	11.8	5.5	1.37	18.4	8.6	2.47	26.5	12.4	4.02
W250x25	6.6	3.2	0.53	11.8	5.7	1.10	18.4	8.9	1.97	26.5	12.8	3.17
W250x24	6.6	3.2	0.35	11.8	5.6	0.75	18.4	8.8	1.39	26.5	12.7	2.31
W250x28	6.6	3.3	0.44	11.8	5.9	0.91	18.4	9.2	1.60	26.5	13.3	2.56
W250x33	6.6	3.5	0.25	11.8	6.2	0.51	18.4	9.7	0.91	26.5	14.0	1.47
W250x39	6.6	3.7	0.20	11.8	6.6	0.41	18.4	10.3	0.71	26.5	14.9	1.13
W250x45	6.6	3.9	0.18	11.8	7.0	0.34	18.4	10.9	0.59	26.5	15.8	0.93
W250x49	6.6	4.1	0.13	11.8	7.3	0.23	18.4	11.4	0.39	26.5	16.3	0.62
W250x58	6.6	4.4	0.10	11.8	7.9	0.19	18.4	12.3	0.32	26.5	17.7	0.50

8.1.2 Wide Flange Shear

Shear forces are assumed to be resisted through the web. Therefore total factored shear is compared against the factored shear capacity of the section web. For a simply supported beam with a UDL the maximum shear and bending do not occur at the same location.

13.4 Shear

13.4.1 Webs of flexural members with two flanges

13.4.1.1 Elastic analysis

The factored shear resistance, V_r , developed by the web of a flexural member shall be taken as

$$V_r = \phi A_w F_s$$

where

A_w = shear area (dw for rolled shapes and hw for girders, $2ht$ for rectangular HSS)

as follows:

a) for unstiffened webs:

i) when $\frac{h}{w} \leq \frac{1014}{\sqrt{F_y}}$:

$$F_s = 0.66 F_y$$

ii) when $\frac{1014}{\sqrt{F_y}} < \frac{h}{w} \leq \frac{1435}{\sqrt{F_y}}$:

$$F_s = \frac{670 \sqrt{F_y}}{(h/w)}$$

iii) when $\frac{h}{w} > \frac{1435}{\sqrt{F_y}}$:

$$F_s = \frac{961 \ 200}{(h/w)^2}$$

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$$1014/(F_y)^{0.5} = 54.2 \quad F_s = 231 \text{ MPa}$$

$$1435/(F_y)^{0.5} = 76.7$$

Maximum ULS factored shear V_f is $0.5*(w_{DL}*l + w_{WL}*l)$

Table 8-7 ULS Factored Shear and Unity Ratios - Case 2 - 1.25DL+1.4WL

Section	Web Thk w (mm)	d (mm)	d/w	F_s (MPa)	V_r (kN)	$V_{f,15'}$ (kN)	$UC_{v,15'}$	$V_{f,20'}$ (kN)	$UC_{v,20'}$	$V_{f,25'}$ (kN)	$UC_{v,25'}$	$V_{f,30'}$ (kN)	$UC_{v,30'}$
W150x22	5.8	152	26	231	183	8.5	0.05	11.3	0.06	14.2	0.08	17.0	0.09
W200x19	5.8	203	35	231	245	8.4	0.03	11	0.05	14	0.06	17	0.07
W200x21	5.0	203	41	231	211	8.5	0.04	11	0.05	14	0.07	17	0.08
W200x22	6.2	206	33	231	266	8.5	0.03	11	0.04	14	0.05	17	0.06
W200x27	5.8	207	36	231	250	8.7	0.03	12	0.05	14	0.06	17	0.07
W200x31	6.4	210	33	231	279	8.8	0.03	12	0.04	15	0.05	18	0.06
W200x36	6.2	201	32	231	259	8.9	0.03	12	0.05	15	0.06	18	0.07
W200x42	7.2	205	28	231	307	9.1	0.03	12	0.04	15	0.05	18	0.06
W200x46	7.2	203	28	231	304	9.3	0.03	12	0.04	15	0.05	19	0.06
W200x52	7.9	206	26	231	338	9.5	0.03	13	0.04	16	0.05	19	0.06
W250x18	4.8	251	52	231	250	8.4	0.03	11	0.04	14	0.06	17	0.07
W250x22	5.8	254	44	231	306	8.5	0.03	11	0.04	14	0.05	17	0.06
W250x25	6.1	257	42	231	326	8.6	0.03	11	0.04	14	0.04	17	0.05
W250x24	5.0	253	51	231	263	8.6	0.03	11	0.04	14	0.05	17	0.07
W250x28	6.4	260	41	231	346	8.7	0.03	12	0.03	14	0.04	17	0.05
W250x33	6.1	258	42	231	327	8.8	0.03	12	0.04	15	0.05	18	0.05
W250x39	6.6	262	40	231	360	9.0	0.03	12	0.03	15	0.04	18	0.05
W250x45	7.6	266	35	231	420	9.2	0.02	12	0.03	15	0.04	18	0.04
W250x49	7.4	247	33	231	380	9.4	0.02	12	0.03	16	0.04	19	0.05
W250x58	8.0	252	32	231	419	9.7	0.02	13	0.03	16	0.04	19	0.05

8.1.2 Local Web Bending

In addition to the failure mechanisms previously outlined, failure caused by local bending in the web of the beam is possible in the absence of stiffeners. This failure mechanism is assessed considering the dead load of the cladding.

Cladding Dead Load $w_{cladding} = 0.98 \text{ kN/m}$ (Considers 15% for connections and 1.4 for load combination a)

The cladding dead load will induce a moment with moment arm $e = d - 2t$. Dead load from the outer flange and web are not included as only the added load will be capable of causing local bending failure.

Failure in this manner is independent of column spacing, therefore the bending moment and factored resistance is computed per meter of girt using elastic analysis

Elastic section modulus of web $S_w = bd^2/6$
Factored bending resistance of web $M_{r,w} = \Phi_s S_w F_y$
Width of section $b = 1 \text{ m}$ (Calculation is per meter of beam)
Height of section $d = w$ (flange thickness)

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Table 8-8 Local Bending about the Web Unity Ratios

Section	Web Thk w (mm)	d (mm)	t (mm)	e (mm)	S_w (mm ³ /m)	$M_{f,w}$ (kNm/m)	$M_{r,w}$ (kNm/m)	$U_{c,w}$
W150x22	5.8	152	6.6	138.8	5606.7	0.14	1.77	0.08
W200x19	5.8	203	6.5	190.0	5606.67	0.19	1.77	0.11
W200x21	5.0	203	6.4	190.2	4166.67	0.19	1.31	0.14
W200x22	6.2	206	8.0	190.0	6406.67	0.19	2.02	0.09
W200x27	5.8	207	8.4	190.2	5606.67	0.19	1.77	0.11
W200x31	6.4	210	10.2	189.6	6826.67	0.19	2.15	0.09
W200x36	6.2	201	10.2	180.6	6406.67	0.18	2.02	0.09
W200x42	7.2	205	11.8	181.4	8640.00	0.18	2.72	0.07
W200x46	7.2	203	11.0	181.0	8640.00	0.18	2.72	0.07
W200x52	7.9	206	12.6	180.8	10401.67	0.18	3.28	0.05
W250x18	4.8	251	5.3	240.4	3840.00	0.24	1.21	0.19
W250x22	5.8	254	6.9	240.2	5606.67	0.23	1.77	0.13
W250x25	6.1	257	8.4	240.2	6201.67	0.23	1.95	0.12
W250x24	5.0	253	6.4	240.2	4166.67	0.23	1.31	0.18
W250x28	6.4	260	10.0	240.0	6826.67	0.23	2.15	0.11
W250x33	6.1	258	9.1	239.8	6201.67	0.23	1.95	0.12
W250x39	6.6	262	11.2	239.6	7260.00	0.23	2.29	0.10
W250x45	7.6	266	13.0	240.0	9626.67	0.23	3.03	0.08
W250x49	7.4	247	11.0	225.0	9126.67	0.22	2.87	0.08
W250x58	8.0	252	13.5	225.0	10666.67	0.22	3.36	0.07

8.2 Deflection Checks

Although there are no live loads applied to the girts, deflections in the horizontal plane due SLS wind load are computed and deflection/length ratios are provided. Girt selection should consider any cladding sensitive to excessive movement. Maximum acceptable SLS deflections are L/180

Girt deflections due to UDL are maximum at mid-span.

Maximum mid-span deflection $\Delta_{\max} = (5 w_{w,sls} l^4)/(384 E I)$

Table 8-9 Maximum SLS Deflection due to Wind

Section	Columns @ 15'		Columns @ 20'		Columns @ 25'		Columns @ 30'	
	$\Delta_{\max,15'}$ (mm)	Δ/L Ratio	$\Delta_{\max,15'}$ (mm)	Δ/L Ratio	$\Delta_{\max,15'}$ (mm)	Δ/L Ratio	$\Delta_{\max,15'}$ (mm)	Δ/L Ratio
W150x22	3.19	L / 1433	10.08	L / 605	24.61	L / 310	51.03	L / 179
W200x19	2.33	L / 1966	7.35	L / 830	17.94	L / 425	37.20	L / 246
W200x21	1.95	L / 2346	6.16	L / 990	15.04	L / 507	31.19	L / 293
W200x22	1.93	L / 2369	6.10	L / 1000	14.89	L / 512	30.88	L / 296
W200x27	1.50	L / 3056	4.73	L / 1289	11.54	L / 660	23.93	L / 382
W200x31	1.23	L / 3720	3.88	L / 1569	9.48	L / 803	19.67	L / 465
W200x36	1.12	L / 4075	3.55	L / 1719	8.66	L / 880	17.95	L / 509
W200x42	0.94	L / 4845	2.98	L / 2044	7.28	L / 1047	15.10	L / 606
W200x46	0.85	L / 5378	2.69	L / 2269	6.56	L / 1162	13.60	L / 672
W200x52	0.73	L / 6243	2.31	L / 2634	5.65	L / 1348	11.72	L / 780
W250x18	1.72	L / 2654	5.45	L / 1119	13.29	L / 573	27.57	L / 332
W250x22	1.34	L / 3424	4.22	L / 1444	10.30	L / 739	21.37	L / 428
W250x25	1.13	L / 4051	3.57	L / 1709	8.71	L / 875	18.06	L / 506
W250x24	1.11	L / 4111	3.52	L / 1734	8.58	L / 888	17.80	L / 514
W250x28	0.96	L / 4738	3.05	L / 1999	7.45	L / 1024	15.44	L / 592
W250x33	0.79	L / 5793	2.49	L / 2444	6.09	L / 1251	12.63	L / 724
W250x39	0.64	L / 7119	2.03	L / 3004	4.96	L / 1538	10.27	L / 890
W250x45	0.54	L / 8423	1.72	L / 3553	4.19	L / 1819	8.69	L / 1053
W250x49	0.55	L / 8363	1.73	L / 3528	4.22	L / 1806	8.75	L / 1045
W250x58	0.44	L / 10342	1.40	L / 4363	3.41	L / 2234	7.07	L / 1293

8.3 Connection Design (In-House)

Typical connections will be designed by connection specialists.

8.4 Connection Design Tables (For Fabricators)

Connections are assumed to be pinned, therefore maximum connection forces consider maximum vertical and horizontal shear loads. Vertical shear (V_y) is caused by self-weight while Horizontal shear (V_h) is caused by wind.

Table 8-10 Connection Design Loading

Section	Columns @ 15'		Columns @ 20'		Columns @ 25'		Columns @ 30'	
	$V_{y,15'}$ (kN)	$V_{h,15'}$ (kN)	$V_{y,20'}$ (kN)	$V_{h,20'}$ (kN)	$V_{y,25'}$ (kN)	$V_{h,25'}$ (kN)	$V_{y,30'}$ (kN)	$V_{h,30'}$ (kN)
W150x22	3.0	5.8	4.0	7.7	5.0	9.6	6.1	11.6
W200x19	2.9	5.8	3.9	7.7	4.9	9.6	5.8	11.6
W200x21	3.0	5.8	4.0	7.7	5.0	9.6	6.0	11.6
W200x22	3.0	5.8	4.0	7.7	5.0	9.6	6.1	11.6
W200x27	3.2	5.8	4.3	7.7	5.4	9.6	6.4	11.6
W200x31	3.4	5.8	4.5	7.7	5.6	9.6	6.7	11.6
W200x36	3.5	5.8	4.7	7.7	5.9	9.6	7.1	11.6
W200x42	3.8	5.8	5.0	7.7	6.3	9.6	7.5	11.6
W200x46	3.9	5.8	5.2	7.7	6.5	9.6	7.8	11.6
W200x52	4.1	5.8	5.5	7.7	6.9	9.6	8.2	11.6
W250x18	2.9	5.8	3.8	7.7	4.8	9.6	5.8	11.6
W250x22	3.0	5.8	4.0	7.7	5.0	9.6	6.1	11.6
W250x25	3.1	5.8	4.2	7.7	5.2	9.6	6.3	11.6
W250x24	3.1	5.8	4.1	7.7	5.2	9.6	6.2	11.6
W250x28	3.2	5.8	4.3	7.7	5.4	9.6	6.5	11.6
W250x33	3.4	5.8	4.6	7.7	5.7	9.6	6.9	11.6
W250x39	3.6	5.8	4.9	7.7	6.1	9.6	7.3	11.6
W250x45	3.9	5.8	5.1	7.7	6.4	9.6	7.7	11.6
W250x49	4.0	5.8	5.3	7.7	6.7	9.6	8.0	11.6
W250x58	4.3	5.8	5.8	7.7	7.2	9.6	8.7	11.6

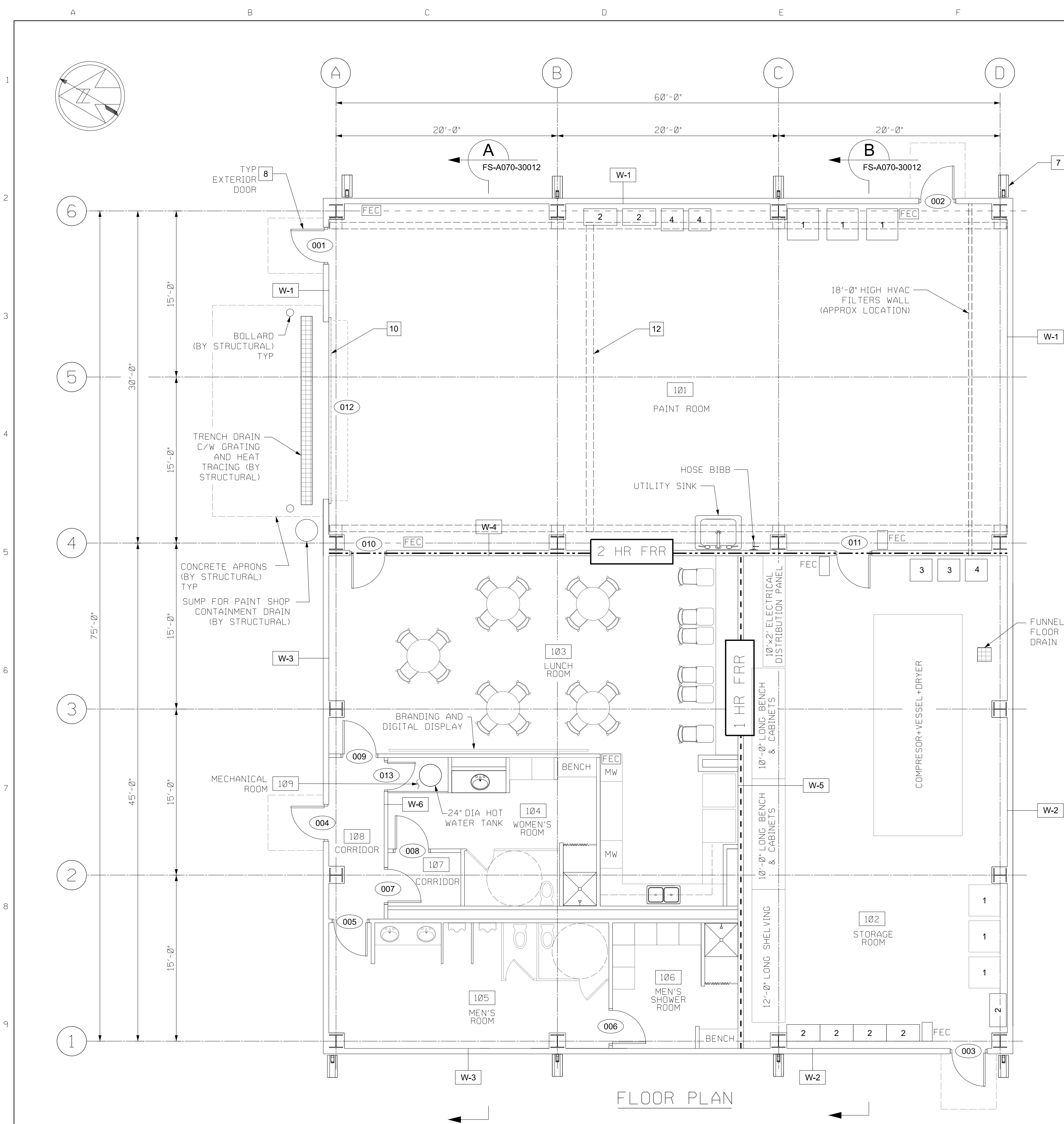


Client: DOW CHEMICAL CANADA ULC
Project No: P2ZU



Designed By: AW .
Checked By: AS .
Date: 27Oct2023 .
Revision: 0 .

9. APPENDIX



WALL ASSEMBLIES (TBC)

EXTERIOR WALL ASSEMBLY SYSTEM

- W-1 - PREFINISHED GALVANIZED METAL PANEL, 22 GAUGE, MORIN (KINGSPAN) EXPOSED VB-3I-S PROFILE 'A' METAL CLADDING OR VICWEST CL7040. (TO BE SIMILAR TO CLADDING PROFILE OF THE EXISTING BUILDINGS ON-SITE)
- STRUCTURAL STEEL GIRTS
 - 8" THICK CONTINUOUS R24 (TBC) MINERAL WOOL INSULATION (MIN DENSITY 96kg/m3) INSTALLED WITH JOINTS STAGGERED
 - AIR/VAPOR RETARDER
 - INTERIOR LINER PRE-COAT WHITE COMPLYING TO NFPA
- W-2 - PREFINISHED GALVANIZED METAL PANEL, 22 GAUGE, MORIN (KINGSPAN) EXPOSED VB-3I-S PROFILE 'A' METAL CLADDING OR VICWEST CL7040. (TO BE SIMILAR TO CLADDING PROFILE OF THE EXISTING BUILDINGS ON-SITE)
- STRUCTURAL STEEL GIRTS
 - 8" THICK CONTINUOUS R24 (TBC) MINERAL WOOL INSULATION (MIN DENSITY 96kg/m3) INSTALLED WITH JOINTS STAGGERED
 - AIR/VAPOR RETARDER
 - FIBREGLASS REINFORCED PLASTIC PANEL

- W-3 - PREFINISHED GALVANIZED METAL PANEL, 22 GAUGE, MORIN (KINGSPAN) EXPOSED VB-3I-S PROFILE 'A' METAL CLADDING OR VICWEST CL7040. (TO BE SIMILAR TO CLADDING PROFILE OF THE EXISTING BUILDINGS ON-SITE)
- STRUCTURAL STEEL GIRTS
 - 8" THICK CONTINUOUS R24 MINERAL WOOL INSULATION (MIN DENSITY 96kg/m3) INSTALLED WITH JOINTS STAGGERED
 - AIR/VAPOR RETARDER
 - 5/8" GYPSUM BOARD (MOISTURE AND MOLD RESISTANT REQUIRED FOR ROOMS 105 AND 106)

INTERIOR WALL ASSEMBLY SYSTEM

2 HOUR FIRE RATED INTERIOR WALL ASSEMBLY - CGC UL U419 C/W FIRE SEALANT AND FIRE STOP

- (SEE MANUFACTURER'S SPECIFICATIONS FOR SIZES, MAXIMUM ALLOWABLE SPANS AND INSTALLATION)
- W-4 - 2 LAYERS 5/8" TYPE 'X' GYPSUM PANEL MOISTURE AND HIGH IMPACT RESISTANT SPEC: SHEETROCK GLASS-MAT MOLD TOUGH VHI FIRECODE X
- 3-5/8" STEEL STUDS, EQ20, 16" OC
 - 1 LAYER FIBREGLASS INSULATION
 - 2 LAYERS 5/8" TYPE 'X' GYPSUM PANEL MOISTURE AND HIGH IMPACT RESISTANT SPEC: SHEETROCK GLASS-MAT MOLD TOUGH VHI FIRECODE X
- (ADD GALVANIZED METAL LINER COMPLYING TO NFPA ON THE PAINT ROOM SIDE AND FRP ON THE STORAGE SIDE)

1 HOUR FIRE RATED INTERIOR WALL ASSEMBLY - CGC UL U419 C/W FIRE SEALANT AND FIRE STOP

- (SEE MANUFACTURER'S SPECIFICATIONS FOR SIZES, MAXIMUM ALLOWABLE SPANS AND INSTALLATION)
- W-5 - 2 LAYERS 5/8" TYPE 'X' GYPSUM BOARD
- 2 1/2" STEEL STUDS 16" OC
 - 1 LAYER 5/8" TYPE 'X' GYPSUM PANEL (MOISTURE AND MOLD RESISTANT REQUIRED ON ROOM 106 SIDE)
 - ADDITIONAL FIBREGLASS REINFORCED PLASTIC PANEL (STORAGE ROOM SIDE)

TYPICAL INTERIOR WALL ASSEMBLY

- W-6 - 5/8" GYPSUM BOARD
- METAL STUDS AT 18" OC
 - 5/8" GYPSUM BOARD (MOISTURE AND MOLD RESISTANT REQUIRED FOR ROOMS 104, 105 AND 106)

KEYNOTE SCHEDULE

1	PREFINISHED GALVANIZED METAL CLADDING
2	HEAVY DUTY PREFINISHED GALVANIZED ROOF
3	RIDGE CAP & FLASHING
4	ICE RAKE
5	HEAVY DUTY PREFINISHED GALVANIZED METAL GUTTER
6	HEAVY DUTY DOWNSPOUT
7	PREF-CAST CONCRETE SPLASH PAD
8	PAINTED AND GALVANIZED INSULATED HOLLOW METAL SINGLE DOOR AND FRAME WITH PANIC HARDWARE
9	SINGLE DOOR CANOPY
10	ELECTRICALLY OPERATED INSULATED ROLL-UP DOOR
11	BUILDING STEEL AND MISCELLANEOUS STEEL FRAMING (BY STRUCTURAL)
12	SMT BRIDGE CRANE (BY CONTRACTOR)

ROOF ASSEMBLY SYSTEM (TBC)

STANDING SEAM METAL ROOF ASSEMBLY

- R-1 - PREFINISHED GALVANIZED STANDING SEAM ROOFING PANEL, 22 GAUGE, MORIN (KINGSPAN) SLR-18-2 PROFILE OR VICWEST TSR. (TO BE SIMILAR TO CLADDING PROFILE OF THE EXISTING BUILDINGS ON-SITE)
- STRUCTURAL STEEL PURLINS
 - 9" THICK CONTINUOUS R30 MINERAL WOOL INSULATION (MIN DENSITY 96kg/m3) INSTALLED WITH JOINTS STAGGERED
 - AIR/VAPOR RETARDER

CEILING TYPES

- C-1 - 2 HOUR FIRE RATED CEILING ASSEMBLY - CGC SA926 (LUNCH ROOM & STORAGE)
- 1" TYPE 'X' GYPSUM BOARD
 - 4" 24ga METAL STUDS @ 2'-0" OC (SEE MANUFACTURER'S SPECIFICATIONS FOR SIZES, MAXIMUM ALLOWABLE SPANS AND INSTALLATION)
 - 3 LAYERS 1/2" FIRE CODE 'C' GYPSUM BOARD
- C-2 - PAINT ROOM
- INTERIOR LINER 18ga PRE-COAT WHITE COMPLYING TO NFPA SUPPORTED BY STRUCTURAL FRAMING (BY STRUCTURAL)

NOTE:

1. FOR LEGEND, GENERAL NOTES, ABBREVIATIONS, DRAWING INDEX AND DRAWING SYMBOLS SEE DRAWING FS-A070-30006.

LEGEND:

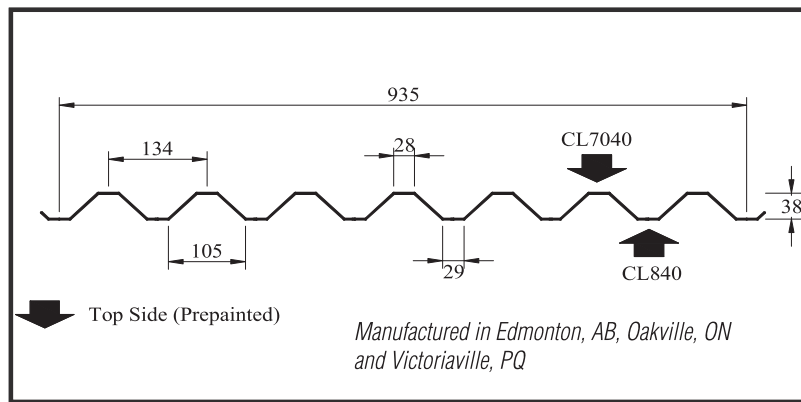
- 1 - FLAMMABLE CABINET - 34"x34"x65"
- 2 - GREY NON-FLAMMABLE LOCKER - 36"x18"x72"
- 3 - BLUE LOCKER - 24"x24"x82"
- 4 - BLAST HOOD CABINET - 24"x24"x72"

LEGEND

-----	2 HOUR FIRE RATING
-----	1 HOUR FIRE RATING

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LIMIT STATES DESIGN



CLADDING CL7040 CL840

Metric

PHYSICAL PROPERTIES

(PER METRE WIDTH)
In accordance with CSA
Specification S136-07

LOAD TABLE

Maximum Specified
Uniformly Distributed
Load in kN/m² (kPa)

Note

1. Properties and loads are based on Grade 230 Steel with a minimum yield stress of 230 MPa, and a maximum stress under Factored loads of 207 MPa.

2. Figures in Row B indicate the load capacity based on strength. Strength capacity B should be checked against [Specified Live Load] + [0.833 x Specified Dead Load].

3. Where cladding is subjected only to wind load, strength values may be increased by 7%.

4. Figures in row D indicate the load capacity based on deflection of 1/180th span. For allowable deflection of 1/90th span, values in Row D can be doubled, but must not exceed the value in Row B. Deflection capacity should be checked against specified Load(s).

5. An * indicates capacity has been reduced to account for web crippling.

Base Steel Nominal Thickness (mm)	Nominal Thickness Z275 Coating (mm)	Mass with Coating (kg/m ²)	Section Modulus		Moment of Inertia (mm ⁴ x 10 ³)	Factored Resistance			
			Midspan (mm ³ x 10 ³)	Support (mm ³ x 10 ³)		Moment (N-m)		Reaction (kN)	
0.46	0.50	5.13	5.44	5.44	113.9	1126.1	1126.1	3.4	5.0
0.61	0.65	6.64	7.75	7.76	152.1	1604.3	1606.3	5.9	8.8
0.76	0.80	8.17	9.77	9.77	190.2	2022.4	2022.4	8.9	13.5
0.91	0.95	9.70	11.70	11.70	228.5	2421.9	2421.9	12.6	19.1
1.22	1.26	12.77	15.47	15.47	304.1	3202.3	3202.3	21.3	32.9

Support Spacing (mm)		1-Span Base Steel Nominal Thickness (mm)					2-Span Base Steel Nominal Thickness (mm)					3-Span Base Steel Nominal Thickness (mm)				
		0.46	0.61	0.76	0.91	1.22	0.46	0.61	0.76	0.91	1.22	0.46	0.61	0.76	0.91	1.22
1200	B	3.8*	5.9	7.5	9.0	11.9	2.2*	3.9*	6.0*	8.5*	11.9	2.5*	4.4*	6.8*	9.6*	14.8
	D	5.7	7.6	9.5	11.5	15.2	13.7	18.3	22.9	27.5	36.6	10.8	14.4	18.0	21.6	28.8
1400	B	3.1	4.4	5.5	6.6	8.7	1.9*	3.4*	5.1*	6.6	8.7	2.2*	3.8*	5.8*	8.2	10.9
	D	3.6	4.8	6.0	7.2	9.6	8.6	11.5	14.4	17.3	23.0	6.8	9.1	11.3	13.6	18.1
1600	B	2.3	3.3	4.2	5.0	6.7	1.7*	2.9*	4.2	5.0	6.7	1.9*	3.3*	5.1*	6.3	8.3
	D	2.4	3.2	4.0	4.8	6.4	5.8	7.7	9.7	11.6	15.4	4.6	6.1	7.6	9.1	12.2
1800	B	1.9	2.6	3.3	4.0	5.3	1.5*	2.6*	3.3	4.0	5.3	1.7*	3.0*	4.2	5.0	6.6
	D	1.7	2.3	2.8	3.4	4.5	4.1	5.4	6.8	8.1	10.8	3.2	4.3	5.3	6.4	8.5
2000	B	1.5	2.1	2.7	3.2	4.3	1.3*	2.1	2.7	3.2	4.3	1.5*	2.7*	3.4	4.0	5.3
	D	1.2	1.6	2.1	2.5	3.3	3.0	4.0	4.9	5.9	7.9	2.3	3.1	3.9	4.7	6.2
2200	B	1.2	1.8	2.2	2.7	3.5	1.2*	1.8	2.2	2.7	3.5	1.4*	2.2	2.8	3.3	4.4
	D	0.9	1.2	1.5	1.9	2.5	2.2	3.0	3.7	4.5	5.9	1.8	2.3	2.9	3.5	4.7
2400	B	1.0	1.5	1.9	2.2	3.0	1.0	1.5	1.9	2.2	3.0	1.3*	1.9	2.3	2.8	3.7
	D	0.7	1.0	1.2	1.4	1.9	1.7	2.3	2.9	3.4	4.6	1.3	1.8	2.3	2.7	3.6
2600	B		1.3	1.6	1.9	2.5		1.3	1.6	1.9	2.5	1.1	1.6	2.0	2.4	3.2
	D		0.7	0.9	1.1	1.5		1.8	2.2	2.7	3.6	1.1	1.4	1.8	2.1	2.8
2800	B		1.1	1.4	1.6	2.2		1.1	1.4	1.6	2.2		1.4	1.7	2.1	2.7
	D		0.6	0.8	0.9	1.2		1.4	1.8	2.2	2.9		1.1	1.4	1.7	2.3
3000	B			1.2	1.4	1.9			1.2	1.4	1.9		1.2	1.5	1.8	2.4
	D			0.6	0.7	1.0			1.5	1.8	2.3		0.9	1.2	1.4	1.8
3200	B			1.1	1.3	1.7			1.1	1.3	1.7		1.0	1.3	1.6	2.1
	D			0.5	0.6	0.8			1.2	1.4	1.9		0.8	1.0	1.1	1.5

In accordance with ongoing efforts to improve our products and their performance, Vicwest reserves the right to change without notice the specifications contained herein.

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VW00153EN01/11



Client: DOW CHEMICAL CANADA ULC
Project No: P2ZU



Designed By: AW .
Checked By: AS .
Date: 27Oct2023 .
Revision: 0 .

9.2 Construction Information



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9.3 Critical Correspondence and Attachments



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9.4 CSA Calculation Checklist



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9.5 Revisions